

ERMA

Equipment Reference Master App

Application design document — use cases, process flows, data model, screens

Big C SuperCenter PCL · BJC Group · Digital Technology Group

Version 2.0 · October 2026 · Scope: MVP

1. Objective

ERMA (Equipment Reference Master App) is the single reference master for equipment maintained by Big C's maintenance-service vendors. Vendors register their equipment models together with a bill of materials (BOM) of key spare parts, including warranty and replacement-part details, so Big C can plan maintenance and claim warranty consistently. ERMA maps each equipment record to the CMS Asset Type and Service Type masters, and publishes equipment information to both the CMS and the Fixed Asset system through scheduled batch jobs.

In scope (MVP)	Out of scope (later phases)
Vendor onboarding (bulk + single), password email, vendor login	Spare-part stock movements, purchasing, reordering
Equipment master, spare-part catalog, BOM with warranty dates	Work orders and maintenance execution (remain in CMS)
Mapping to CMS asset / service type	Asset depreciation (remains in Fixed Asset system)
Batch sync: CMS → ERMA, ERMA → CMS, ERMA → Fixed Asset	Real-time API integration

2. Actors and use cases

Actors: **BigC Admin** (bulk vendor onboarding), **BigC Editor** (single vendor registration and deletion), **Vendor** (owner of equipment, spare-part and BOM data), and three external systems — **Email service**, **CMS** (source of asset/service type masters, receiver of equipment data) and **Fixed Asset** (receiver of equipment data).

ID	Use case	Actor	Trigger / notes
UC1	Bulk upload vendor list	BigC Admin	CSV/Excel; row-level validation, error report; includes UC4
UC2	Register new vendor	BigC Editor	Web form; includes UC4
UC3	Delete / deactivate vendor	BigC Editor	Soft delete; blocks login, keeps history
UC4	Send password email	System	Temp password generated and emailed for every new vendor
UC5	Vendor login	Vendor	Emailed credentials; forced change on first login
UC6	Create / edit equipment master	Vendor	Tag, brand, model, sub model, spec, location, warranty
UC7	Maintain spare parts and BOM	Vendor	Pick from catalog or add part; qty, critical flag, S/N, warranty
UC8	Map / remap asset & service type	Vendor	Select from CMS-mirrored masters (UC9)
UC9	Batch: CMS → ERMA asset & service types	System (batch)	Nightly; refreshes read-only mapping masters
UC10	Batch: ERMA → CMS equipment	System (batch)	Nightly; records with record_status = queued
UC11	Batch: ERMA → Fixed Asset equipment	System (batch)	Nightly; records with record_status = queued

2.1 Use case diagram

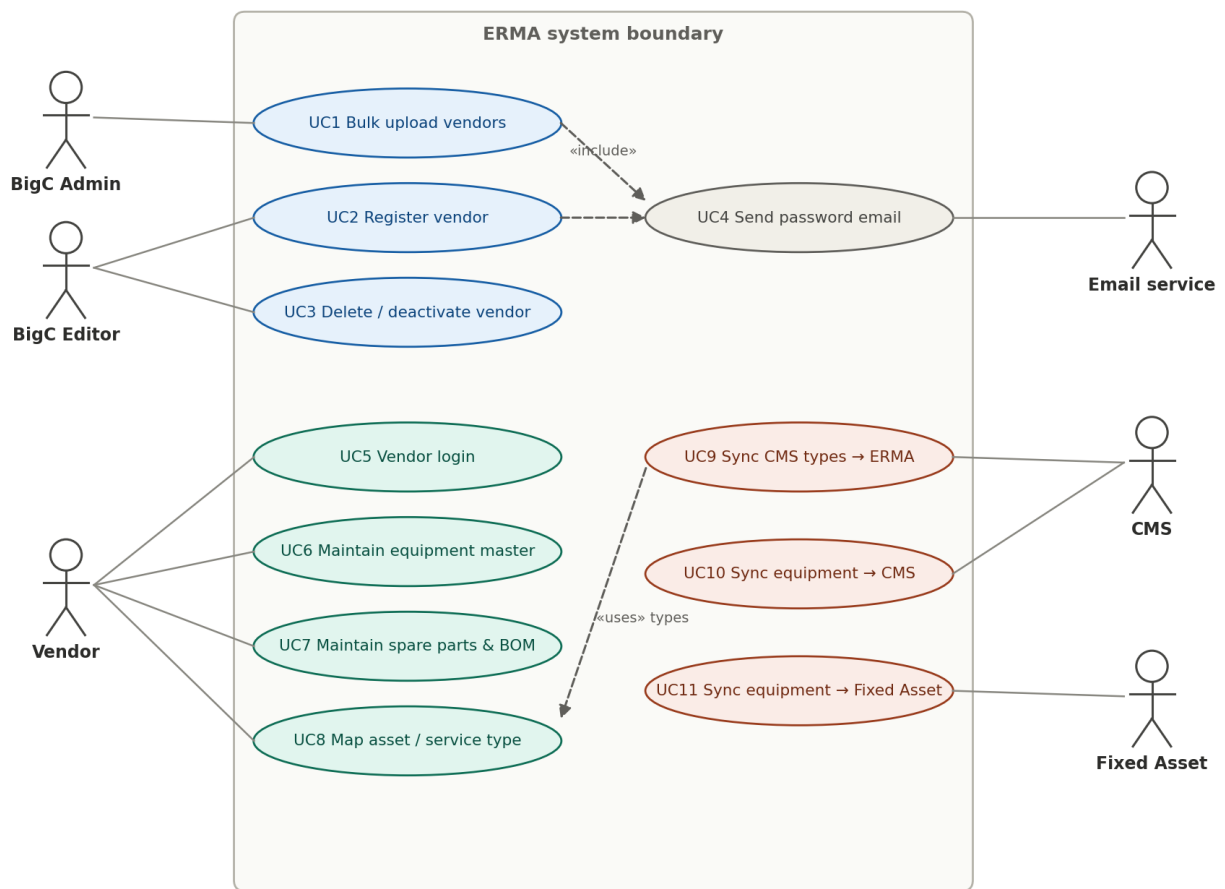


Figure 1. ERMA use case diagram.

3. Process flows

3.1 Vendor onboarding (UC1–UC5)

Bulk and single registration share one validation path. Duplicate tax ID or email is rejected; the remaining rows create SUPPLIER records in *pending* status, receive a hashed temp password, and are emailed. The account becomes *active* after the forced password change at first login.

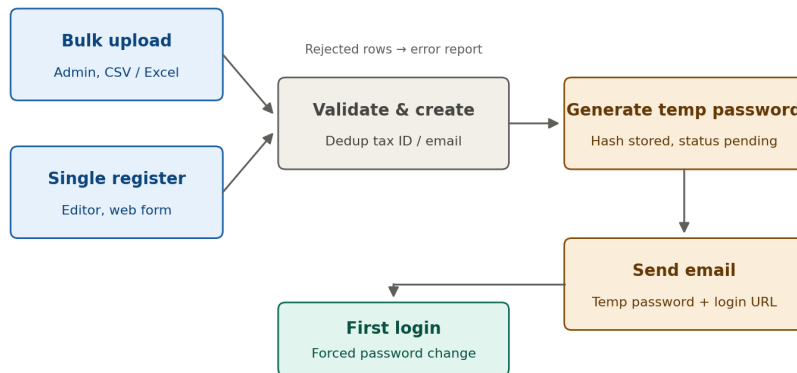


Figure 2. Vendor onboarding flow.

3.2 Equipment and BOM creation (UC6–UC8)

The vendor enters the equipment header, builds the BOM by picking spare parts from the catalog (or creating a new catalog part), sets quantity, critical flag, serial number and warranty start, maps the CMS asset and service type, then saves. A valid save sets record_status to *queued* for both outbound batches.

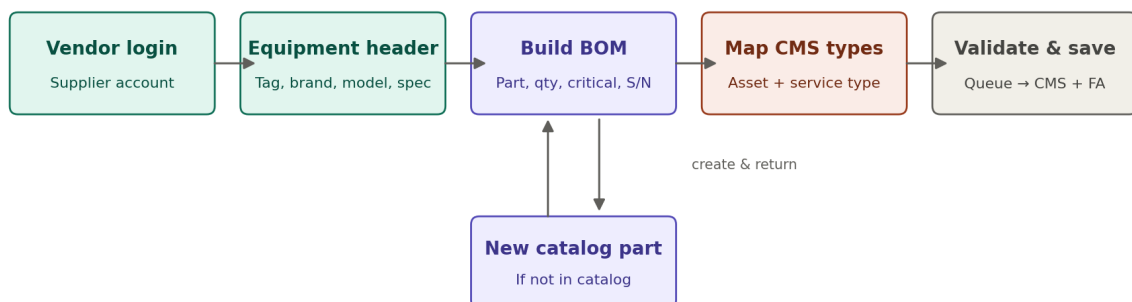


Figure 3. Equipment and BOM creation flow.

3.3 CMS and Fixed Asset batch sync (UC9–UC11)

The inbound job runs first so that mappings use current CMS types. The two outbound jobs then send queued equipment to CMS and to the Fixed Asset system. Each record and each target writes one SYNC_LOG row.

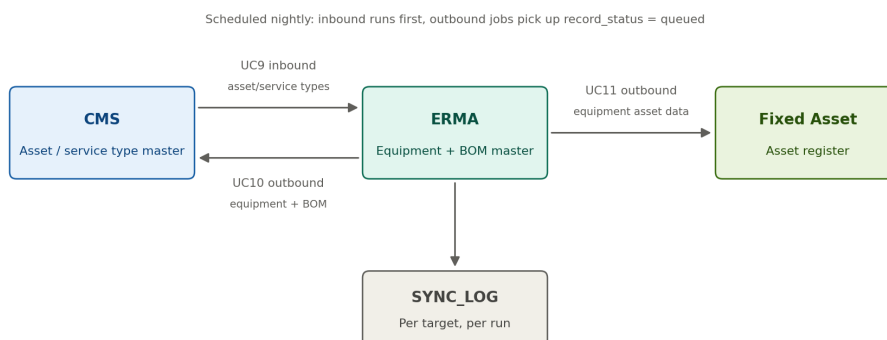


Figure 4. Batch sync flow.

4. Equipment status lifecycle

EQUIPMENT carries two separate status fields. **status** (from the source model) is the operational state — Active, In Repair, Decommissioned. **record_status** (added by ERMA) is the master-data lifecycle below, which drives the batch jobs.

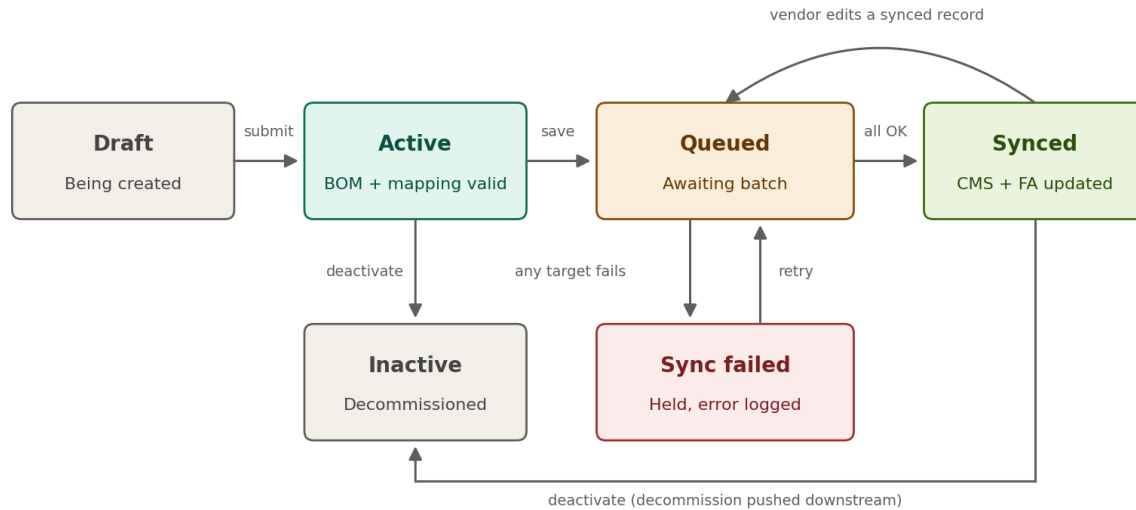


Figure 5. record_status transitions.

record_status	Meaning	Entered from	Exit trigger
draft	Header saved; BOM or mapping incomplete	New record	Submit with valid BOM + mapping
active	Complete and valid master record	draft	Save → queued; deactivate → inactive
queued	Waiting for nightly outbound batches	active, synced (on edit), sync_failed (retry)	All targets OK → synced; any fails → sync_failed
synced	Latest version accepted by CMS and Fixed Asset	queued	Vendor edit → queued; deactivate → inactive
sync_failed	At least one target rejected; error in SYNC_LOG	queued	Retry (manual or next run) → queued
inactive	Decommissioned; excluded from new BOM work	active, synced	Terminal (reactivation by Editor only)

5. Data model

The model follows the attached Excel (Equipment, Spare part, BOM, Supplier) and adds the entities ERMA needs to run: BIGC_USER for Admin/Editor login, LOCATION referenced by equipment, the two CMS mapping masters, and SYNC_LOG. SPARE_PART is a catalog; BOM is the many-to-many link between EQUIPMENT and SPARE_PART and holds instance data (serial number, warranty dates). The Excel Supplier is the ERMA vendor and is extended with login fields.

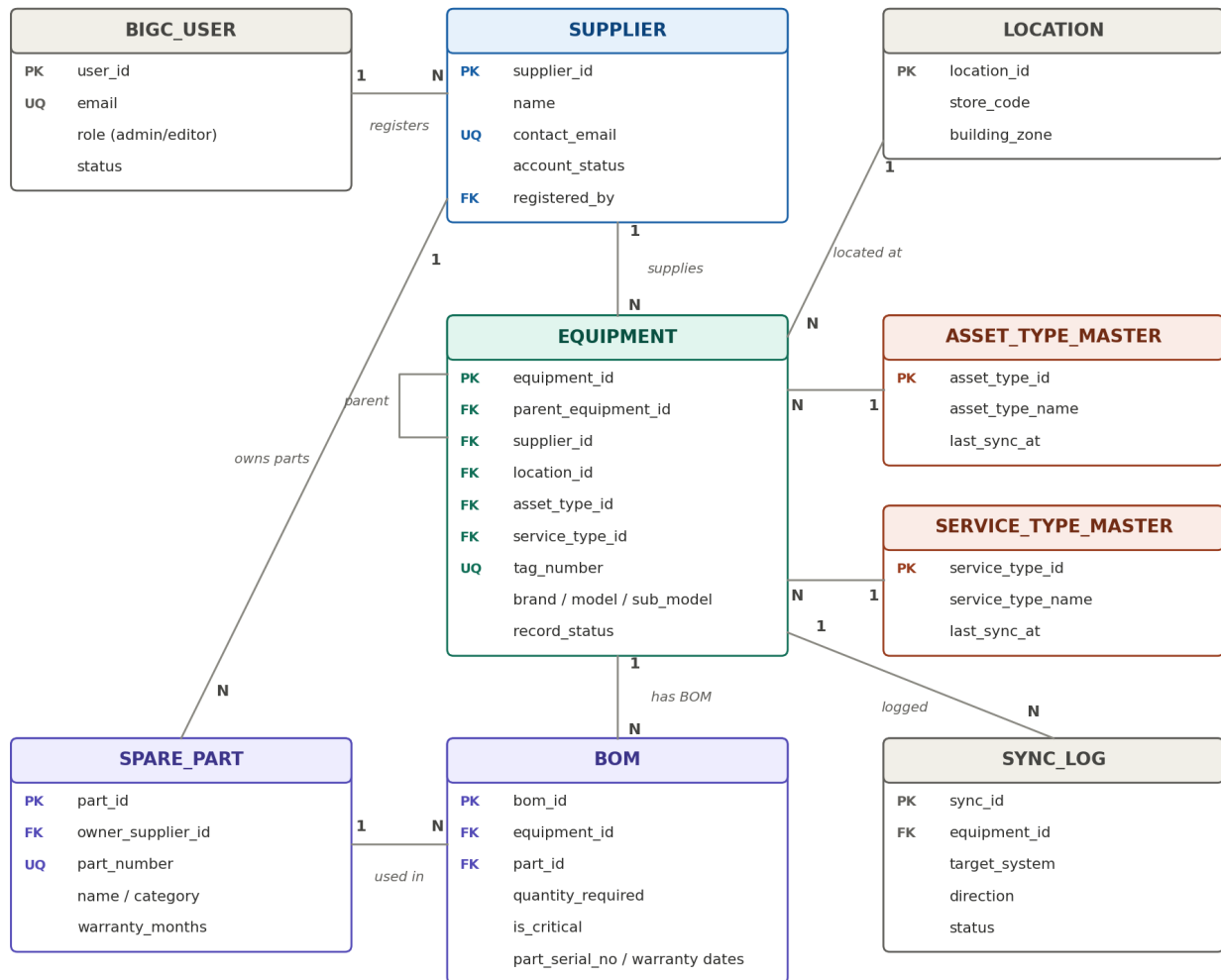


Figure 6. Entity relationship diagram (key fields only; full detail in 5.1).

5.1 Field-level data dictionary

One row per column. **Source:** *Excel* = from the attached model (names normalised to snake_case); *ERMA* = added for this design.

Entity	Column	Data type	Constraints	Description	Source
SUPPLIER	supplier_id	INT / UUID	PK	Unique identifier	Excel
SUPPLIER	name	VARCHAR	NOT NULL	Supplier company name	Excel
SUPPLIER	lead_time_days	INT		Average delivery time for reordering	Excel
SUPPLIER	contract_price	DECIMAL		Agreed purchasing cost	Excel
SUPPLIER	tax_id	VARCHAR	UNIQUE	Tax registration number; duplicate check	ERMA
SUPPLIER	contact_email	VARCHAR	UNIQUE, NOT NULL	Login and password-delivery email	ERMA
SUPPLIER	contact_phone	VARCHAR		Contact phone	ERMA
SUPPLIER	password_hash	VARCHAR		Hashed password (never plaintext)	ERMA
SUPPLIER	must_change_password	BOOLEAN	DEFAULT TRUE	Force change at first login	ERMA
SUPPLIER	account_status	VARCHAR	NOT NULL	pending / active / inactive	ERMA
SUPPLIER	registration_type	VARCHAR		bulk / single	ERMA
SUPPLIER	registered_by	UUID	FK → BIGC_USER	Admin/Editor who registered	ERMA
SUPPLIER	registered_at	DATETIME		Registration timestamp	ERMA
BIGC_USER	user_id	UUID	PK	Unique identifier	ERMA
BIGC_USER	email	VARCHAR	UNIQUE, NOT NULL	Corporate login (SSO recommended)	ERMA
BIGC_USER	display_name	VARCHAR		User name	ERMA
BIGC_USER	role	VARCHAR	NOT NULL	admin / editor	ERMA
BIGC_USER	status	VARCHAR		active / inactive	ERMA
LOCATION	location_id	INT	PK	Physical plant / building / zone	ERMA
LOCATION	store_code	VARCHAR	NOT NULL	Big C store (Oracle) code	ERMA
LOCATION	building_zone	VARCHAR		Building, floor or zone	ERMA
EQUIPMENT	equipment_id	INT / UUID	PK	Unique identifier	Excel
EQUIPMENT	parent_equipment_id	INT / UUID	FK (self)	Parent in asset tree (nullable)	Excel
EQUIPMENT	tag_number	VARCHAR	UNIQUE	Physical asset tag / barcode	Excel
EQUIPMENT	name	VARCHAR	NOT NULL	Equipment name, e.g. Chiller Unit A	Excel
EQUIPMENT	category	VARCHAR		e.g. Chiller, Lift, Air conditioner	Excel
EQUIPMENT	location_id	INT	FK → LOCATION	Physical location	Excel
EQUIPMENT	status	VARCHAR		Operational: Active / In Repair / Decommissioned	Excel
EQUIPMENT	install_date	DATE		Commissioning date	Excel
EQUIPMENT	brand	VARCHAR		e.g. Hitachi, Samsung	Excel
EQUIPMENT	model	VARCHAR		e.g. HT-348-EX	Excel
EQUIPMENT	sub_model	VARCHAR		e.g. OUT10	Excel
EQUIPMENT	equipment_type	VARCHAR		e.g. Ceiling type	Excel
EQUIPMENT	short_spec	VARCHAR		Short specification, e.g. 300 W	Excel
EQUIPMENT	full_spec	JSON		Full specification as JSON	Excel
EQUIPMENT	warranty_months	INT		Equipment warranty, months from install	Excel
EQUIPMENT	warranty_end_date	DATE	Calculated	install_date + warranty_months	ERMA
EQUIPMENT	criticality_score	VARCHAR		High / Medium / Low	Excel
EQUIPMENT	supplier_id	INT / UUID	FK → SUPPLIER	Owning vendor	Excel
EQUIPMENT	asset_type_id	VARCHAR	FK → ASSET_TYPE_MASTER	CMS asset type	ERMA
EQUIPMENT	service_type_id	VARCHAR	FK → SERVICE_TYPE_MASTER	CMS service type	ERMA
EQUIPMENT	record_status	VARCHAR	NOT NULL	draft / active / queued / synced / sync_failed / inactive	ERMA
EQUIPMENT	cms_sync_status	VARCHAR		Last CMS outbound result	ERMA
EQUIPMENT	fa_sync_status	VARCHAR		Last Fixed Asset outbound result	ERMA
EQUIPMENT	created_at / updated_at	DATETIME		Audit timestamps	ERMA
EQUIPMENT	updated_by	VARCHAR		Last editor (supplier or BigC user)	ERMA
SPARE_PART	part_id	INT / UUID	PK	Unique part identifier	Excel

Entity	Column	Data type	Constraints	Description	Source
SPARE_PART	part_number	VARCHAR	UNIQUE	Manufacturer part number or internal SKU	Excel
SPARE_PART	name	VARCHAR	NOT NULL	Part description, e.g. Ball Bearing 6205	Excel
SPARE_PART	category	VARCHAR		Consumable / Repairable / Capital Spare	Excel
SPARE_PART	part_type	VARCHAR		Compressor, fan, motor, mother board, power supply	ERMA
SPARE_PART	model	VARCHAR		Spare part model, if any	Excel
SPARE_PART	warranty_months	INT		Warranty, months from install / replacement	Excel
SPARE_PART	unit_of_measure	VARCHAR		Each, Meters, Liters, Box	Excel
SPARE_PART	standard_cost	DECIMAL		Moving average or baseline cost	Excel
SPARE_PART	min_stock_level	INT		Reorder threshold (safety stock)	Excel
SPARE_PART	max_stock_level	INT		Maximum storage capacity	Excel
SPARE_PART	owner_supplier_id	INT / UUID	FK → SUPPLIER	Vendor that created the part	ERMA
BOM	bom_id	INT / UUID	PK	Unique identifier	Excel
BOM	equipment_id	INT / UUID	FK → EQUIPMENT	Parent equipment	Excel
BOM	part_id	INT / UUID	FK → SPARE_PART	Catalog part	Excel
BOM	quantity_required	INT	NOT NULL	Number used in this assembly	Excel
BOM	is_critical	BOOLEAN		True if failure halts operation	Excel
BOM	part_serial_no	VARCHAR		Installed part S/N	ERMA
BOM	warranty_start_date	DATE		Install / replacement date; defaults to equipment install_date	ERMA
BOM	warranty_end_date	DATE	Calculated	warranty_start_date + SPARE_PART.warranty_months	ERMA
ASSET_TYPE_MASTER	asset_type_id	VARCHAR	PK	CMS asset type key (read-only mirror)	ERMA
ASSET_TYPE_MASTER	asset_type_name	VARCHAR		Description	ERMA
ASSET_TYPE_MASTER	is_active	BOOLEAN		Active in CMS	ERMA
ASSET_TYPE_MASTER	last_sync_at	DATETIME		Last inbound refresh	ERMA
SERVICE_TYPE_MASTER	service_type_id	VARCHAR	PK	CMS service type key (read-only mirror)	ERMA
SERVICE_TYPE_MASTER	service_type_name	VARCHAR		Description	ERMA
SERVICE_TYPE_MASTER	is_active	BOOLEAN		Active in CMS	ERMA
SERVICE_TYPE_MASTER	last_sync_at	DATETIME		Last inbound refresh	ERMA
SYNC_LOG	sync_id	UUID	PK	Log row identifier	ERMA
SYNC_LOG	batch_run_id	UUID		Groups rows of one run	ERMA
SYNC_LOG	equipment_id	INT / UUID	FK → EQUIPMENT	Null for inbound master refresh	ERMA
SYNC_LOG	target_system	VARCHAR	NOT NULL	CMS / FIXED_ASSET	ERMA
SYNC_LOG	direction	VARCHAR	NOT NULL	inbound / outbound	ERMA
SYNC_LOG	status	VARCHAR	NOT NULL	success / failed	ERMA
SYNC_LOG	error_message	VARCHAR		Target error detail	ERMA
SYNC_LOG	run_at	DATETIME		Execution time	ERMA

6. Screen inventory and wireframes

The MVP needs an estimated **6 screens**. Spare-part catalog maintenance is handled inside the equipment detail screen (add from catalog / create part) for the MVP.

#	Screen	Actor	Use cases	Key elements
1	Login + first-login password change	All	UC5	Email/password, forced change, forgot password
2	Vendor bulk upload	BigC Admin	UC1, UC4	Template download, file upload, validation result, error report
3	Vendor management	BigC Editor	UC2, UC3, UC4	Search, register form, deactivate, resend password
4	Equipment list	Vendor	UC6	Search by tag/brand/model, record_status filter, new equipment
5	Equipment detail	Vendor	UC6, UC7, UC8	Header, warranty, CMS mapping, BOM grid, add from catalog
6	Sync monitor	Admin, Editor, Vendor (own)	UC9–UC11	KPIs per target, batch log, retry failed

6.1 Vendor management (screen 3)

Vendor (supplier) management

Bulk upload

+ Register vendor

Search name, tax ID or email

Status: All ▼

Supplier	Contact email	Lead time	Status	Registered
Siam Cool Air	contact@siamcool.co.th	7 days	Active	Bulk · 02 Sep ×
Thai Lift Service	info@thailift.com	14 days	Pending	Single · 15 Sep ×
Cold Chain Eng.	ops@coldchain.co.th	10 days	Inactive	Bulk · 28 Jul ×

Pending = temp password emailed, first login not yet done

Figure 7. Vendor management wireframe.

6.2 Equipment detail (screen 5)

Equipment master · TAG-CH-0001

Queued

Save

Tag number

NAME

Brand

Model

TAG-CH-0001

Chiller Unit A

Carrier

30XA-500

Sub model

Equipment type

Location

Install date

OUT10

Air-cooled

Store 111125 · Roof

01 Sep 2026

Short specification

Warranty (months)

Warranty end (calc.)

Air-cooled screw chiller x 500 kW

24

01 Sep 2028

Criticality

Parent equipment

CMS asset type

CMS service type

High ▾

— ▾

Refrig. - Chiller ▾

PM - Cooling ▾

Bill of materials (BOM)

+ Add from catalog

Part no.	Part name	Qty	Critical	Serial no.	Warranty end
CMP-2201	Screw compressor	2	Yes	SN-88310	01 Sep 2028
FAN-0450	Condenser fan motor	6	—	SN-10021	01 Mar 2027
PCB-0093	Control board	1	Yes	SN-55902	—

Warranty end = warranty start + part warranty months (read-only)

Figure 8. Equipment detail wireframe.

6.3 Sync monitor (screen 6)

Sync monitor

Target: All ▾ Last 7 days ▾

Synced to CMS

1,204

Synced to FA

1,198

Queued

38

Failed

5

Batch job log

Target	Direction	Equipment	Status	Run time	
CMS	Outbound	TAG-CH-0001	Synced	03 Oct 01:00	
Fixed Asset	Outbound	TAG-AC-0120	Failed	03 Oct 01:30	Retry
CMS	Inbound	Asset/service types	Synced	03 Oct 00:30	

Figure 9. Sync monitor wireframe.

7. Implementation notes

Data model

- SPARE_PART is a reusable catalog and BOM is the equipment-to-part link. Serial number and warranty start/end sit on BOM because they belong to the installed part, not the catalog item.
- warranty_end_date (EQUIPMENT and BOM) is calculated on save from the start date and warranty_months, and is read-only in the UI so it cannot drift.
- Excel normalisation: column names converted to snake_case (Brand → brand, Sub model → sub_model); typo supplier_id corrected to supplier_id.
- criticality_score is typed INT in the Excel but holds High / Medium / Low. Use VARCHAR with those values, or INT 1–3 with a lookup — decide before build.
- part_number is UNIQUE in the Excel. With many vendors, confirm whether uniqueness is global or per owner_supplier_id.
- Keep status (operational) and record_status (master-data lifecycle) separate; only record_status drives the batch jobs.

Security

- Store only password hashes; the emailed password is a one-time temporary credential with forced change and an expiry.
- Vendors see and edit only their own SUPPLIER, EQUIPMENT and SPARE_PART rows (row-level filter on supplier_id / owner_supplier_id).
- Vendor delete is a soft delete (account_status = inactive); equipment history and SYNC_LOG stay intact.

Integration

- ASSET_TYPE_MASTER and SERVICE_TYPE_MASTER are read-only mirrors overwritten by UC9; types retired in CMS are flagged is_active = false, not deleted, so existing mappings stay valid.
- Run order: inbound (UC9) first, then outbound (UC10, UC11). Outbound uses idempotent upsert keyed on equipment_id / tag_number so reruns are safe.
- A record moves to synced only when both CMS and Fixed Asset succeed; per-target results are kept in cms_sync_status, fa_sync_status and SYNC_LOG.
- Confirm with the Fixed Asset team whether it returns an asset number; if so, add a fixed_asset_no column on EQUIPMENT.

Environments

- **[NON-PRODUCTION]** Build and test all three batch jobs against CMS and Fixed Asset test instances with sample vendor data first; use a test mailbox for password emails.
- **[PRODUCTION — affects live systems]** UC10 and UC11 write to live CMS and Fixed Asset. Go-live needs a signed-off field mapping, a dry-run file review, and a rollback plan. The first bulk vendor upload sends real emails and cannot be recalled — validate the file before running it.